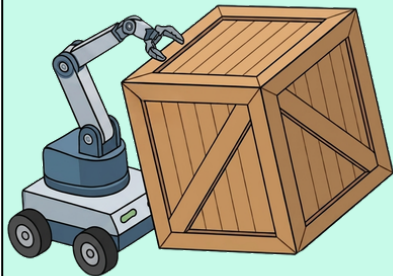


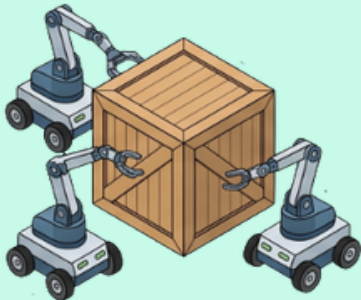
CHALLENGE:

Large object transport



- Single Robot Limitations
- Object too Heavy/Bulky
- Instability risks

COOPERATIVE MULTI ROBOT SYSTEM



- Formation Control
- Coordinated Motion
- Shared Load

COORDINATION WITH HUMAN

HUMAN-IN-THE-LOOP

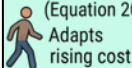
ADAPTIVE COORDINATION PROTOCOL

5.1 PROACTIVE FEEDBACK
via Cost Monitoring

NMPC
PREDICTIVE COST (J)

If $J < J_{\text{thresh}}$ →
SIGNAL: PROCEED

If $J \geq J_{\text{thresh}}$ ⚠
SIGNAL: SLOW DOWN
(Equation 20)



Adapts
rising cost

5.2 REACTIVE SAFETY
SUPERVISION

(Distance Monitoring)
DISTANCE MONITORING
($D_{hr} = \|p_h - p_{obj}\|$)

If $D_{hr} \geq D_{\text{stop}}$ ⛔
SIGNAL: STOP

If $D_{hr} < D_{\text{start}}$ →
SIGNAL: PROCEED
(Equation 21)

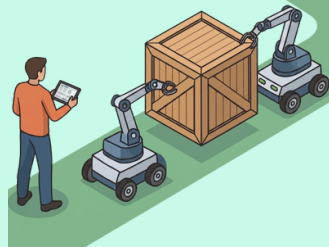
Stop
and waiting



OUTCOME

SUCCESSFUL AND SAFE
TRANSPORTATION

Destination 



- Successful object delivery
- Increased Load capacity
- Obstacle avoidance